Experimental Considerations of CVD Diamond Film Measurements using Time Domain Thermoreflectance

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ABSTRACT

Diamond has the highest known thermal conductivity of any known bulk material, but the properties of synthetic diamond films often fall far short of this high level. The DARPA program Thermal Transport in Diamond Films for Electronics Thermal Management brings together researchers from five universities to comprehensively characterize the thermal transport and material properties of CVD diamond thin films in an effort to better how to further improve the thermal transport properties and understand how accurately these properties can be measured using time domain thermoreflectance and Raman spectroscopy. Here we summarize the results of the thermal measurements of diamond conducted via time domain thermoreflectance (TDTR) using two different systems and discuss some difficulties of accurately measuring the thermal conductivity of micron-thick anisotropic films that often have high surface roughness. We also report that in certain cases the thermal conductivity and thermal boundary conductance of CVD diamond films has been improved to the point of making them highly attractive for thermal management of high power electronic devices.

KEY WORDS: Diamond; Thermal Conductivity; Thermal Management; Time Domain Thermoreflectance

INTRODUCTION

Among the many attractive properties of diamond is that it has the highest thermal conductivity of any bulk material, >2000 W/m-K at room temperature [1]. Synthetic diamond grown via chemical vapor deposition (CVD) has been studied for some time [2], and high thermal conductivity (500-2000 W/m-K) can be achieved in thick films of several hundred microns [3-5]. Despite the ability to grow thick diamond films with high thermal conductivity, the boundary resistance between the diamond and the growth substrate can remain a bottleneck to heat flow [6], along with the diamond in the first few microns near the substrate where the thermal conductivity is much lower [7]. The small size of the seed diamonds on the substrate leads to nano-scale grains near the interface that increase in size as the film grows. Increasing the heat conduction across the boundary and within the first few microns of diamond is critical for electronics applications. Removing this bottleneck will help unlock the full potential of diamond substrates in use of electronics thermal managements for ultra-high power density applications. A recent theoretical study has shown that if these interfacial problems are eliminated, heat flux removal on the order of 50 kW/cm² is achievable with diamond substrates when combined with advanced fluidic cooling [8].

A variety of techniques have been used to measure the thermal conductivity of diamond including flash diffusivity [9, 10], 3-omega [11], steady-state four probe [12], suspended electrical heaters [13], Raman thermography [14], time domain thermoreflectance (TDTR) [7, 15]. The boundary resistance between diamond and adjacent layers has been measured using Raman [6], TDTR [16], and nanosecond reflectance [17].

Here we discuss measurements of diamond films conducted as a part of the DARPA Thermal Transport in Diamond Films for Electronics Thermal Management program and discuss some of the difficulties associated with making accurate thermal conductivity measurements of thin diamond films. We focus here on the measurements made using TDTR, although the program also involves Raman thermography.

Diamond Thermal Conductivity Measurements

Time domain thermoreflectance has become one of the most popular methods to measure thermal conductivity over the past decade owing to its multi-layer analytical solution [18], fine spatial resolution, potential for high accuracy, and ability to resolve thin film properties and interfaces. It has been used to measure thermal conductivity spanning a range of < 0.1 W/m-K [19] to greater than 1,000 W/m-K and various samples such as polymer thin films, superlattices, graphene interfaces, liquids, etc.

TDTR is a pump-probe photothermal technique that employs an ultra-fast laser to heat a sample and then measure its temperature response over a few ns. The pump (heating) pulse is modulated in the range of 1-10 MHz, which not only controls the penetration depth of heat into the sample, but also allows a lock-in amplifier to extract the temperature response of the surface with a much higher signal to noise ratio. The probe (temperature sensing) pulse travels through a mechanical stage that can delay the arrival of the probe relative to the pump pulse over a range of 0.1 to several ns, which is how a temperature decay curve is captured. The maximum delay time is 7 ns for the GT system and 3.8 ns for the Stanford system. The temperature of the sample is sensed by measuring the change in reflectivity of a thin metal layer on the sample

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surface. For the systems used in this work the probe beam is at a wavelength of 800 nm (GT) and 1064 nm (Stanford) while the pump wavelength is 400 nm (GT) and 532 nm (Stanford). Al is used as the transducer due to a high thermoreflectance coefficient at that wavelength [20]. The thickness of the Al film is nominally 90 nm and measured using picosecond acoustics [21] with the GT system. The pulse with is several hundred fs on the GT system and 9 ps on the Stanford system. The repetition rates of both laser oscillators are near 80 MHz. In addition to normal internal calibration checks, a single crystal Si sample was measured on both systems and the thermal conductivity measurements were both within 5% of 145 W/m-K.

CVD diamond films present several challenges to TDTR measurements due to some of the non-idealities of the samples. Typical diamond samples are rough, inhomogeneous, and anisotropic. High surface roughness can reduce the collected light, and introduce additional error into the theoretical analysis due to non-planar layers. In some cases the samples can be polished to reduce the surface roughness, however the inhomogeneous and isotropic nature of the films must still be handled. For anisotropic materials, there are exact analytical solutions for anisotropy in 2D [22] and 3D [23] however this makes the determination of thermal conductivity and thermal boundary resistance more difficult with the additional unknown property. Typically using a single modulation frequency it is possible to extract two unknown properties from the sample. There is always an unknown boundary resistance between the Al transducer in the sample, which means in most cases the measurement can extract the layer thermal conductivity. However, for diamond, samples the in-plane and through-plane thermal conductivity is different meaning that additional measurements are needed to extract both properties; this can be done through changing the heating frequency, laser spot size, or sample geometry.

An additional difficulty is in the determination of the heat capacity of the diamond; CVD diamond can have a range of values for heat capacity depending on the growth quality and the amount of non-diamond carbon (NDC) present in the sample [24]. In the case of a thin-diamond film (< 5 µm), the measurement will penetrate fully to the underlying substrate, which makes the measurement sensitive to the diamond-substrate boundary resistance as well. In this case there may be a total of five unknown parameters: Al-diamond TBR, diamond through-plane conductivity, diamond in-plane conductivity, diamond heat capacity, and diamond-substrate TBR. Even when combining on-substrate and suspended membrane measurements, extracting all unknown parameters accurately is difficult.

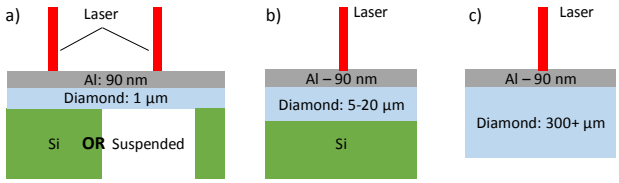


Figure 1: Sample configurations for diamond thermal conductivity measurements. a) Thin diamond films on Si and suspended b) Diamond films on Si c) Thick freestanding diamond films.

We measure diamond properties in several different configurations, as shown in Figure 1. We can measure the in-plane thermal conductivity (κ_r) of thin diamond films (~ 1 µm thick) using suspended membranes or the through-plane thermal conductivity (κ_z) and thermal boundary resistance (TBR) when the film is on an Si substrate. We also measure the through-plane thermal conductivity on thicker diamond on Si (5-20 µm thick) and on very thick freestanding films (>300 µm thick). Each of these sample configurations presents unique challenges, but here we will focus on measurement of κ_r on thin suspended diamond membranes and measurement of κ_z on thick freestanding diamond films. In this paper, we will discuss some of the difficulties associated with making such measurements using TDTR, compare some results between two different TDTR systems, and estimate the uncertainty.

TDTR Sensitivity Analysis

We analyze the sensitivity of the TDTR measurement to the properties of interest to assess the ability to perform good measurements under various experimental conditions. It is also important to acknowledge the sampling volume of TDTR with respect to the sample dimensions; thermal conductivity measurements using TDTR are often taken as the bulk material properties despite the fact that a much smaller volume is being sampled. For high quality single crystal semiconductors this assumption is valid, however for more defective and heterogeneous materials such as CVD diamond this assumption is only an approximation. It is well known that CVD diamond often has a large gradient in grain size from nucleation to free growth surface and that this may correspond to a gradient in thermal conductivity as well [11, 15]. In addition local variations in amount of non-diamond carbon (NDC) [25], grain size [26], or surface roughness [27] may impact the local measurement of thermal conductivity as well. The sampling depth of the measurement is controlled by the heating frequency, typically ~1-10 MHz. Figure 2 shows how the penetration depth of the thermal wave changes with modulation (heating) frequency for different diamond thermal conductivity values. For low conductivity and high heating frequencies, the measurement may only sample 1-2 µm into the film, while for high conductivity and low frequencies this depth can approach 20 µm. Even for this case, a sampling depth of 20 µm can be much smaller than the film thickness that can be over 300 µm.

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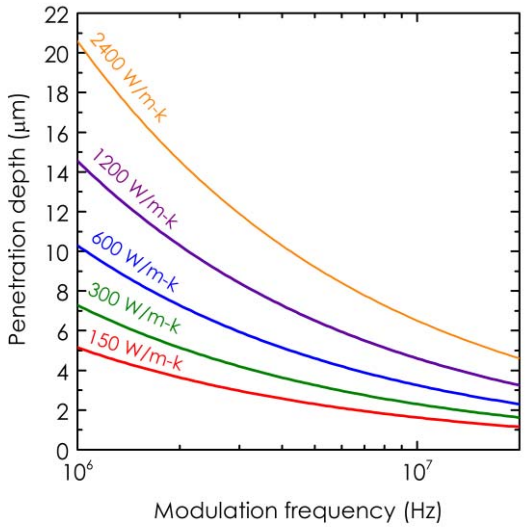


Figure 2. Penetration depth of TDTR sampling versus heating (modulation) frequency for various diamond thermal conductivity values.

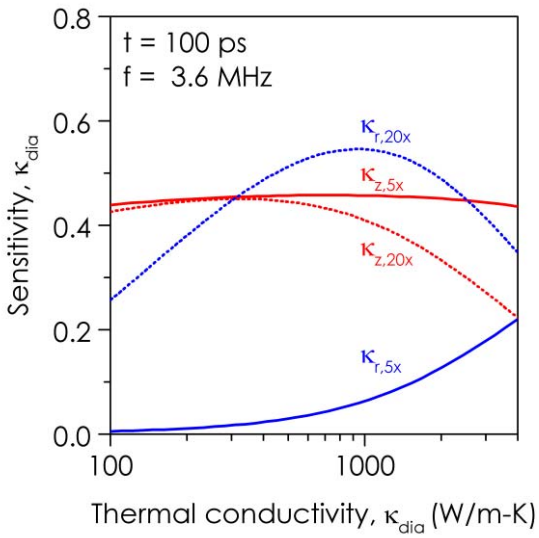


Figure 3. Sensitivity to through-plane (κ_z) and in-plane (κ_r) thermal conductivity at t = 100 ps, f = 3.6 MHz for 5x and 20 objective.

Depending on the specific optics, the TDTR laser spot sizes typically range from 4 to 100 μm in diameter; properly choosing the appropriate spot size in conjunction with a heating frequency can significantly alter whether the measurement is that of a local area (or even grain) or closer to an average value of the diamond film. We study of the effect of grain size on thermal conductivity in other work [28], as well as the effect of varying thermal conductivity [7, 29]; in this work we focus on choosing the proper experimental conditions for the measurements and how different inputs to the model contribute to uncertainty.

To properly determine the appropriate measurement parameters and sample configuration for accurate property determination we perform a sensitivity analysis. The sensitivity is defined as [30]:

$$S_p = \frac{dR/R}{dP/P} \qquad (1)$$

where $R = -V_{in}/V_{out}$ (the ratio of the in phase to the out of phase voltage of the lock-in amplifier) and p is the parameter of interest. Figure 3 shows the sensitivity of κ_z and κ_r for a thick freestanding diamond film measured at 3.6 MHz with a delay time of 100 ps. Previous work on high conductivity diamond using TDTR used the in-phase signal (V_{in}) to fit thermal properties rather than the ratio, ($-V_{in}/V_{out}$), due to the low Vout signal level owing to a combination of high conductivity and also high applied pressure (> 1 GPa) [31]. In this work we are measuring at atmospheric pressure and use high laser power to ensure the V_{out} signal is large enough allow standard ratio fitting.

For this simple analysis we use isotropic diamond thermal conductivity so that both parameters vary equally. Because the diamond layer is thermally thick, the through-plane thermal conductivity is relatively constant with thermal conductivity when using the 5x objective lens (10/40 μm beam diameter). At low thermal conductivity values (<500 W/m-K) the sensitivity to κ_r is minimal indicating 1D conduction normal to the film. When the thermal conductivity is greater than 1000 W/m-K there is significant sensitivity to κ_r and κ_z indicating 2D radial conduction. When using the smaller spot size of the 20x objective (4/10 μm beam diameter) both κ_z and κ_r are important over the entire conductivity range, however κ_r becomes more significant for higher conductivity. Ideally, the TDTR measurement should be performed with the biggest spot size possible when the interest is in κ_z; when attempting to measure κ_r of thick diamond films, there will always be a contribution of κ_z. This will require two separate measurements using a large spot and high frequency to determine κ_z and then a small spot and low frequency to determine κ_r.

We apply this same principle to the measurement of κr and κ_z of 1 μm diamond films on an Si substrate. For the sensitivity analysis we assume that κ_z = 100 W/m-K, κ_r = 50 W/m-K and $TBR_{Si\text{-}dia}$ = 50 GW/m²-K. The sensitivity to κ_z will be high at f = 12 MHz with a 5x objective (Figure 4a), while the sensitivity to κ_r will be high for f = 1 MHz, 20x objective (Figure 4b).

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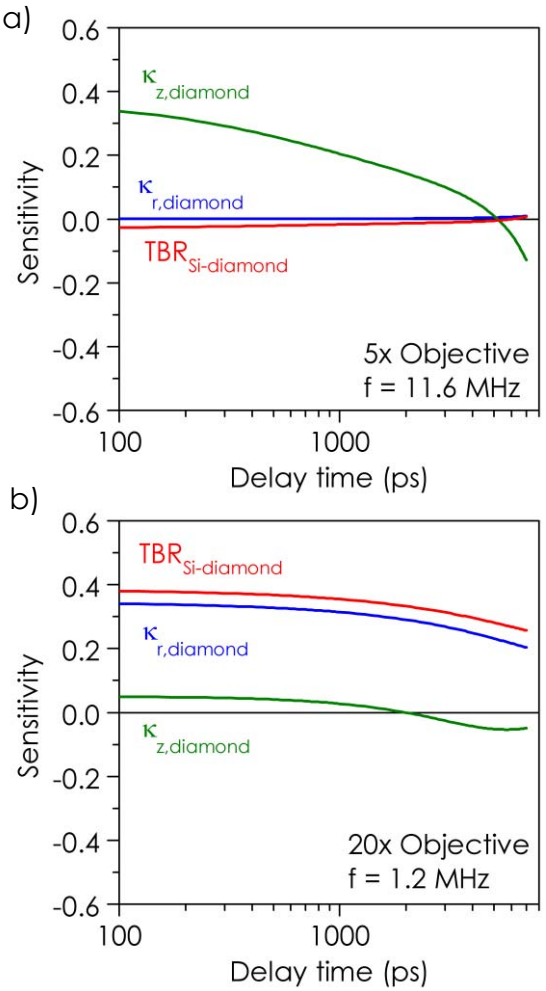


Figure 4. Sensitivity for 1 µm diamond film on Si a) 12 MHz modulation frequency and 5x objective b) 1 MHz modulation frequency and 20x objective

The main difficulty with isolating κ_r and κ_z by frequency and spot size for a 1 µm diamond film is there is still an unknown TBR between the diamond and substrate that is unknown, but important in the TDTR measurement (Figure 4b). By created a suspended membrane, not only will the extra TBR be removed, but the heat will be forced to conduct radially, increasing the sensitivity to κ_r and decreasing the sensitivity to κ_z.

The thermal boundary resistance between the Al transducer and the diamond layer can negatively impact the ability to accurately resolve the diamond thermal conductivity if the boundary resistance is too high. As can be seen in Figure 5, for TBR values of greater than 10 m²-K/GW, the sensitivity to $\kappa_{diamond}$ is reduced. However, for lower TBR values the sensitivity is relatively independent of both TBR and the $\kappa_{diamond}$ value.

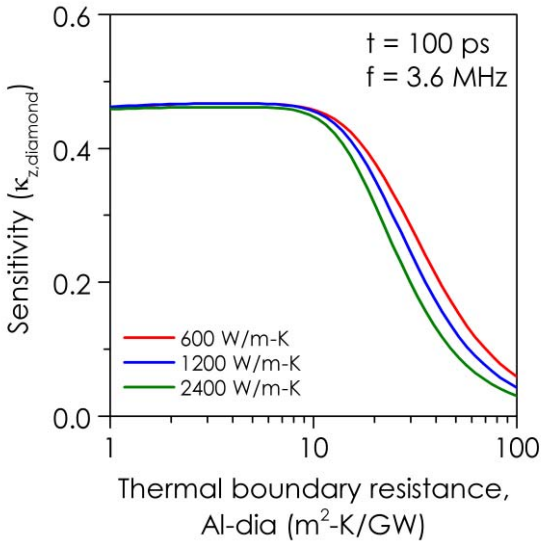


Figure 5. Sensitivity to diamond thermal conductivity versus Al-diamond thermal boundary resistance at t = 100 ps, f = 3.6 Mz.

RESULTS & DISCUSSION

Thin Diamond Films & Suspended Membranes

For a suspended membrane, a single frequency measurement can determine κ_r and $TBR_{Al\text{-}dia}$ because the sensitivity to κ_z is low and there is no TBR between diamond and a growth substrate. This works well if the heat capacity is well known, but this can vary significantly for synthetic diamond [32]. We use a Monte Carlo uncertainty algorithm to estimate the contribution of different model inputs to the total uncertainty of the in-plane thermal conductivity measurement. Figure 6 shows the relative contribution to uncertainty to each of the model input parameters. The input parameters are heat capacity of diamond (C_dia), through-plane Al thermal conductivity (kz_Al), in-plane Al thermal conductivity (kr_Al), heat capacity of Al (C_Al), Al thickness (t_Al), pump radius (r_pump) and probe radius (r_probe). The uncertainty for each of these parameters varies from 2% to 10% as noted in parentheses in the figure.

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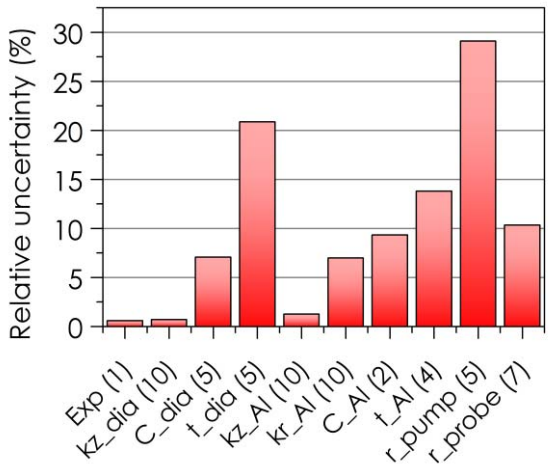


Figure 6. Uncertainty due to different TDTR model inputs a) 1 µm suspended diamond membrane.

The three main contributions to uncertainty for the suspended membrane measurement are the pump spot size, the thickness of the diamond layer, and the thickness of the Al layer. The Al thickness is determined by picosecond acoustics [21] and the laser spot sizes are determined by a beam profiler. The diamond thickness is measured by ellipsometry. This plot shows that using a grower's nominal diamond film thickness or inaccurate methods of measuring spot size can result in significant errors in determine in-plane conductivity.

Figure 7 shows how the uncertainty changes as the in-plane thermal conductivity value changes for a heating frequency of 1.2 MHz. The uncertainty is highest for $\kappa_r < 30$ W/m-K, where the measurement sensitivity is quite low to κ_r; with such low conductivity very little heat is conducted laterally outside of the initial spot during the 7 ns measurement. When κ_r reaches a value of ~70 W/m-K the uncertainty plateaus around 12% with only minimal further improvements. The 1 µm diamond samples measured have in-plane thermal conductivity ranging from 50 to 100 W/m-K resulting in an uncertainty of 11-15% from the model. As discussed in our related paper, the local variability in diamond thermal conductivity can add considerably more uncertainty to the measurement [33].

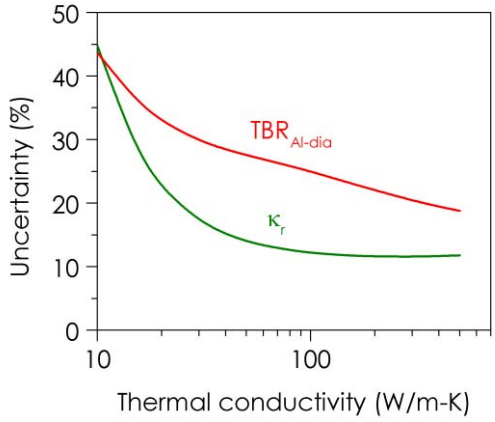


Figure 7. Projected uncertainty in TDTR measurements: In-plane thermal conductivity (κ_r) and thermal boundary resistance between Al and diamond ($TBR_{Al\text{-}dia}$) in 1 µm suspended membrane (1.2 MHz).

As mentioned in the sensitivity analysis multiple frequencies are needed to determine κz on thin films (<3 µm). Figure 8 shows an example of simultaneously fitting four frequencies on 1 µm diamond film on Si to measure κ_z, $TBR_{Al\text{-}dia}$, and $TBR_{dia\text{-}Si}$.

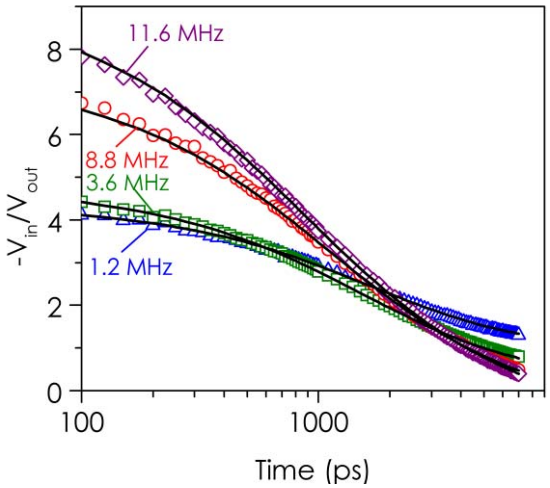


Figure 8. Multi-frequency fitting of 1 µm diamond film on Si. Simultaneously fitting 4 frequencies to extract through-plane thermal conductivity (κ_z = 123 W/m-K), Al-diamond TBR (6.5 m²-K/GW) and diamond-Si TBR (14.7 m^2-K/GW).

The uncertainty of κ_z in such a measurement is highly dependent upon the value of κ_z; the uncertainty when κ_z = 120 W/m-K is approximately 25%; for higher conductivity films the uncertainty increases further. The upper limit of resolvable thermal conductivity for a 1 µm diamond film on Si is in the range of 200 W/m-K. This can be extended higher by increasing the diamond film thickness or increasing the conductivity of the underlying substrate (e.g. switching from Si to SiC).

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Thick Free Standing Diamond

The uncertainty for measurements of thick freestanding diamond is more straightforward in terms of model inputs; however, there are additional experimental issues that can contribute to error. Larger laser spot sizes and thick diamond films remove the high sensitivity to these two parameters. Instead, thick diamond can have issues with high surface roughness and accurate focusing. The TDTR thermal model assumes the layers are completely planar and cannot account for surface roughness. In practicality, the surface roughness of thick diamond can be 50 nm rms or greater. In addition to the loss of signal due to scattered light, high surface roughness makes the model a less accurate depiction of the true sample geometry. Investigation of the error caused by surface roughness is beyond the scope of this work, but it is an important issue to consider. It should be noted that both TDTR systems used in this work are two-color (i.e. pump and probe beams are different wavelengths of light), so they do not suffer from the same depolarization issues that would affect measurements of rough samples using polarized TDTR systems [27].

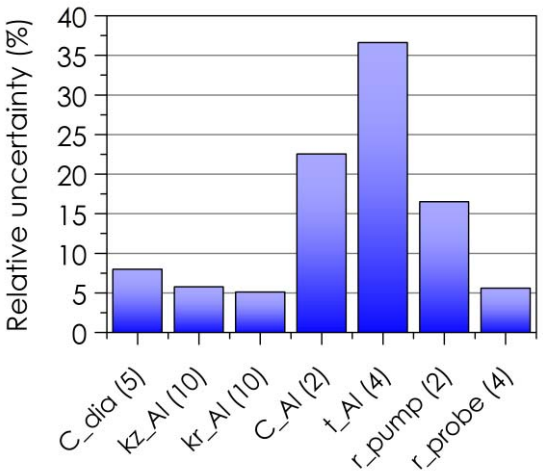


Figure 9. Uncertainty due to different TDTR model inputs a) 1 µm suspended diamond membrane b) Bulk freestanding diamond film

For thick diamond samples, the main contribution to model uncertainty are the heat capacity and thickness of the Al layer. The Al heat capacity value is taken from literature, and this parameter deserves further consideration in future work.

Figure 10 shows the experimental data and theoretical fits for three diamond samples of varying thermal conductivity. The samples have through-plane conductivity values of 650, 1470, and 2200 W/m-K, respectively.

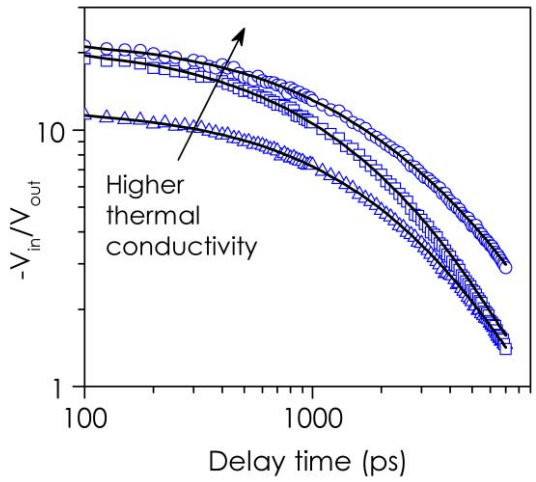


Figure 10. Experimental data for three freestanding films at 3.6 MHz, κ_z = 650 W/m-K (triangles), κ_r = 1470 W/m-K (squares), and κ_z = 2200 W/m-K (circles).

The ratio of $-V_{in}/V_{out}$ increases for higher conductivity samples. The difference is slope is largely due to changes in the Al-diamond TBR. The ratio of the two high conductivity samples is approximately 20 at a delay time of 100 ps. If the in-phase signal (V_{in}) is 100 µV this means V_{out} would be only -5 µV. Keeping the background noise level low is critical for making accurate high conductivity measurements. Longer averaging times will also minimize the contributions from small fluctuations in V_{out}.

We compare the measurements on high conductivity free-standing measured by the two TDTR systems with a laser flash measurement (Figure 11). The uncertainty increases for higher conductivity, but the agreement over the range of 600 to 2200 W/m-K is better than 10% for all samples. The TDTR error bars include contributions from theoretical uncertainty and spot-to-spot variability (based on five measurements). The flash measurement samples a much larger area (d ~ 1 mm) compared with the TDTR measurement (d ~ 40 µm), which is contributing to the large TDTR error bars. While TDTR has often been used to measure the thermal conductivity of materials such as SiC [34], Si, and GaN [35-37], we believe this is the first report of TDTR through-plane thermal conductivity measurements in the range of 500-2000 W/m-K verified with secondary measurements. For the samples with κ_z > 1000 W/m-K we find that the measurement variability is reduced as the laser spot sizes are increased, indicating that small errors in finding the focal point can become significant with ultra-high conductivity samples. To perform the measurement of 2200 W/m-K with the GT system we had to replace the 5x objective with a different lens increasing the pump diameter from 40 to 70 µm. The upper limit of conductivity that can be measured with TDTR is still under investigation, but it appears that with careful attention to experimental details and model input parameters samples above 1000 W/m-K can be measured with this technique.

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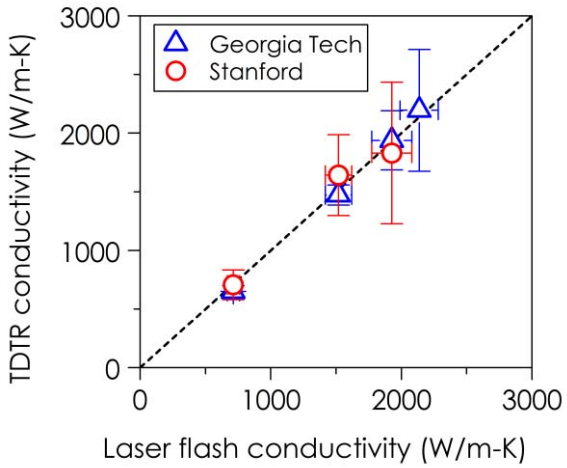


Figure 11. Comparison of thermal conductivity measurements from TDTR and laser flash for freestanding diamond films [38].

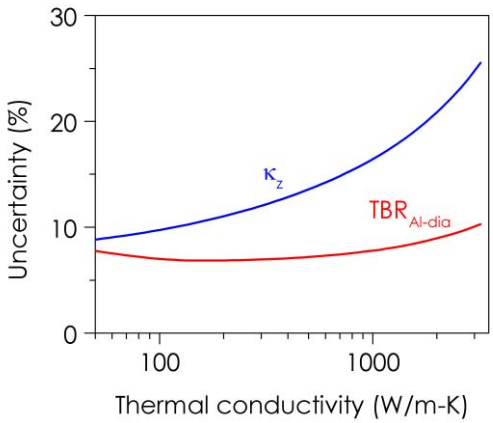


Figure 12. Projected uncertainty in TDTR measurements: Through-plane thermal conductivity (κ_z) and thermal boundary resistance between Al and diamond ($TBR_{Al\text{-}dia}$) in freestanding diamond film membrane.

Figure 12 shows the projected uncertainty for thick diamond samples as a function of through-plane thermal conductivity. The model uncertainty is approximately 8% for $\kappa_z = 50$ W/m-K and 25% for $\kappa_z = 2400$ W/m-K. It is interesting to note that the trend of uncertainty with thermal conductivity is opposite for κ_r on thin membranes and κ_z on thick samples. The uncertainty for κ_r decreases as the value gets larger for membranes, while the uncertainty for κ_z increases for thick samples. For in-plane measurements, there needs to be adequate lateral conduction to sense a different in the surface temperature, but for through-plane measurements, the conductivity should not be so great to minimize the temperature response of the surface.

Conclusions

In this paper, we have discussed some of the experimental difficulties in measuring different types of diamond samples using time domain thermoreflectance. If appropriate care is taken to minimize the effect of certain experimental and theoretical sources of error, then accurate measurements of through-plane and in-plane thermal conductivity can be made in many situations. The properties of CVD diamond including high surface roughness, anisotropy, heterogeneity, and sometimes ultra-high conductivity make these materials challenging for any experimental thermal characterization. As engineers move towards incorporating diamond into high power electronics for heat dissipation we can use tools such as TDTR to study some of these non-idealities to help develop better diamond growth and more accurate thermophysical properties for thermal modeling.

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